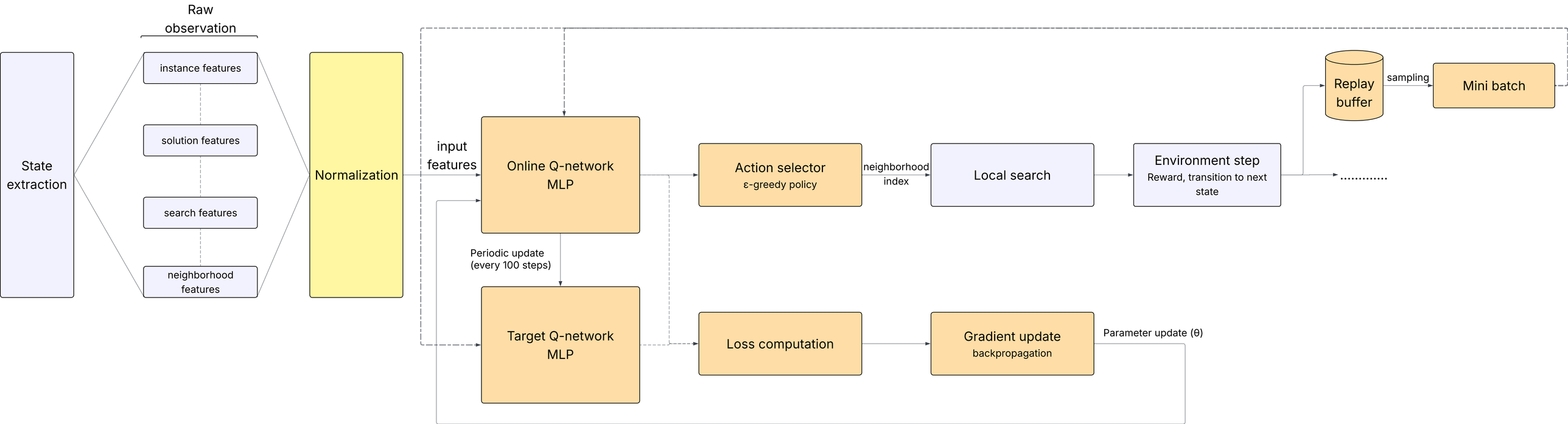
Raw observation
State extraction
instance features
solution features
search features
neighborhood features
Normalization
input features
Online Q-network MLP
Action selector
ε-greedy policy
neighborhood index
Local search
Environment step
Reward, transition to next state
Replay buffer
sampling
Mini batch
.....
Periodic update (every 100 steps)
Target Q-network MLP
Loss computation
Gradient update
backpropagation
Parameter update (θ)